

T0 Theory: The Casimir Scale as Cosmic Link

Casimir Effect, CMB and Hubble Constant in a ξ -Power Ladder

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Contents

.1	Background: Three Apparently Unconnected Phenomena	3
	Casimir Effect and Vacuum Structure (Doc. 091)	3
	H_0 Ratio (Doc. 165)	3
	The Open Question	3
.2	Derivation of the Bridge Formula	3
	L_ξ from T_{CMB}	3
	Multiplication of the Two Relations	4
	The Exponent $41/4$ as Sum of Two Known Exponents	4
.3	The Complete ξ -Power Ladder	4
	Five Scales in One Ladder	4
	Step Structure	5
	The Geometric Mean	5
.4	Physical Interpretation	5
	Prefactors and Their Meaning	5
	On the Decomposition of the Exponent $41/4$	5
.5	Experimental Testability	6
	Casimir Precision Measurements	6
	H_0 Measurement as Indirect Casimir Test	6
.6	Complete Numerical Chain	7
.7	Summary	7

Abstract

This document shows that the Casimir effect, the cosmic microwave background (CMB) and the Hubble constant H_0 are not independent phenomena on different scales, but manifestations of the same ξ geometry at different power levels. The Casimir vacuum scale $L_\xi \approx 100 \mu\text{m}$ (Doc. 091) and the H_0 ratio $\hbar H_0 / m_e c^2 = (\pi/2) \cdot \xi^{10}$ (Doc. 165) are linked by an exact algebraic bridge formula:

$$L_\xi \cdot \frac{H_0}{c} = \frac{\pi}{2} \cdot \left(\frac{15}{\pi^2} \right)^{1/4} \cdot \frac{m_e c^2}{k_B T_{\text{CMB}}} \cdot \xi^{41/4} \quad (1)$$

The exponent $41/4$ is identical to the one in Doc. 026 ($E_H = E_0 \cdot \xi^{41/4}$) and is composed of the H_0 exponent 10 and the Casimir exponent $1/4$. Five seemingly unconnected scales – from the sub-Planck length to the Hubble length – form a gapless ξ -power ladder with step spacings $\Delta n = 1; 5.77; 2.18; 7.78$. Since L_ξ itself is determined from the measured CMB energy density and ξ , the Casimir–CMB connection is a statement about scales, not measurement evidence; the exponent 10 of the H_0 relation is calibrated to the measured H_0 (Doc. 165).

.1 Background: Three Apparently Unconnected Phenomena

Casimir Effect and Vacuum Structure (Doc. 091)

Doc. 091 shows that the energy density of the cosmic microwave background is described by the T0 vacuum as:

$$\rho_{\text{CMB}} = \frac{\xi \hbar c}{L_\xi^4} \quad (2)$$

Numerically, the characteristic vacuum length scale comes out as $L_\xi \approx 100 \mu\text{m}$; at this scale the Casimir energy density is $\pi^2/(720 \xi) \approx 102.8$ times the CMB energy density:

$$\rho_{\text{Casimir}}(d = L_\xi) = \frac{\pi^2 \hbar c}{720 L_\xi^4} = \frac{\pi^2}{720 \xi} \rho_{\text{CMB}} \quad (3)$$

Substituting (2), equation (3) reproduces the standard Casimir formula exactly – ξ cancels out. The Casimir and CMB energy densities become equal only at $d = (\pi^2 \hbar c / (720 \rho_{\text{CMB}}))^{1/4} = (\pi^2 / (720 \xi))^{1/4} L_\xi = 319 \mu\text{m}$ with $\rho_{\text{CMB}} = 4.17 \times 10^{-14} \text{ J/m}^3$.

H₀ Ratio (Doc. 165)

Doc. 165 shows:

$$\frac{\hbar H_0}{m_e c^2} = \frac{\pi}{2} \xi^{10} \quad (4)$$

This yields $H_0 \approx 66.8 \text{ km/s/Mpc}$ – within 0.9% of the Planck value 67.4 km/s/Mpc . The exponent 10 is not derived forward here, but determined numerically from the measured H_0 values (Doc. 165).

The Open Question

The two results – (2) and (4) – were developed independently of each other and refer to entirely different physical phenomena. Are they really independent? Or are they two aspects of the same ξ structure?

.2 Derivation of the Bridge Formula

L_ξ from T_{CMB}

Combining the Casimir vacuum relation (2) with the Stefan–Boltzmann radiation formula $\rho_{\text{CMB}} = (\pi^2/15)(k_B T_{\text{CMB}})^4/(\hbar c)^3$ gives L_ξ as a function of T_{CMB} :

$$L_\xi = \left[\frac{15 \xi}{\pi^2} \right]^{1/4} \frac{\hbar c}{k_B T_{\text{CMB}}} \quad (5)$$

Numerically: $L_\xi = 100.24 \mu\text{m}$ (deviation from the value obtained directly from the measured ρ_{CMB} : 0.11%).

Multiplication of the Two Relations

We multiply L_ξ from (5) by H_0/c from (4):

$$\begin{aligned} L_\xi \cdot \frac{H_0}{c} &= \left[\frac{15\xi}{\pi^2} \right]^{1/4} \frac{\hbar c}{k_B T_{\text{CMB}}} \cdot \frac{\pi}{2} \frac{m_e c^2 \xi^{10}}{\hbar c} \\ &= \frac{\pi}{2} \left[\frac{15}{\pi^2} \right]^{1/4} \frac{m_e c^2}{k_B T_{\text{CMB}}} \cdot \xi^{10+1/4} \\ &= \frac{\pi}{2} \left[\frac{15}{\pi^2} \right]^{1/4} \frac{m_e c^2}{k_B T_{\text{CMB}}} \cdot \xi^{41/4} \end{aligned} \quad (6)$$

This is formula (1). The derivation is exact – no approximation, no free parameters.

Bridge Formula Casimir $\leftrightarrow$ H_0

$$L_\xi \cdot \frac{H_0}{c} = \underbrace{\frac{\pi}{2} \left(\frac{15}{\pi^2} \right)^{1/4}}_{K_{\text{geo}}=1.7441} \cdot \underbrace{\frac{m_e c^2}{k_B T_{\text{CMB}}}}_{=2.176 \times 10^9} \cdot \xi^{41/4} \quad (7)$$

Numerical verification: $L_\xi H_0/c = 7.241 \times 10^{-31}$ (direct and algebraic evaluation identical to within $10^{-14}\%$), with $m_e c^2/(k_B T_{\text{CMB}}) = 510\,998.95 \text{ eV} / (8.6173 \times 10^{-5} \text{ eV/K} \cdot 2.7255 \text{ K}) = 2.176 \times 10^9$.

The Exponent 41/4 as Sum of Two Known Exponents

$$\frac{41}{4} = \underbrace{10}_{H_0 \text{ exponent (Doc. 165)}} + \underbrace{\frac{1}{4}}_{\text{Casimir exponent}} \quad (8)$$

The same exponent appears in Doc. 026 as $E_H = E_0 \cdot \xi^{41/4}$. This means that the decomposition $41/4 = 10 + 1/4$ is not accidental but reflects the physical structure – 10 describes the coupling strength from the electron mass to the Hubble scale, $1/4$ describes the Casimir scaling of the vacuum energy.

.3 The Complete ξ -Power Ladder

Five Scales in One Ladder

We compute all relevant scales as ξ powers of the Planck length ℓ_P :

Table 1: The ξ -power ladder: five length scales and their ξ exponents

Scale	Physical meaning	Length [m]	n : scale $\sim \xi^n \cdot \ell_P$
$L_0 = \xi \cdot \ell_P$	Sub-Planck, T0 lattice	2.16×10^{-39}	$n = +1.0$
ℓ_P	Planck length	1.62×10^{-35}	$n = 0.0$
$\bar{\lambda}_e$	Compton wavelength	3.86×10^{-13}	$n = -5.77$
L_ξ	Casimir vacuum scale	1.00×10^{-4}	$n = -7.95$
c/H_0	Hubble length	1.38×10^{26}	$n = -15.72$

Step Structure

Table 2: Step spacings in the ξ -power ladder

Transition	Δn	Factor
$L_0 \rightarrow \ell_P$	-1.0	$1/\xi$
$\ell_P \rightarrow \bar{\lambda}_e$	-5.77	$\xi^{-5.77}$
$\bar{\lambda}_e \rightarrow L_\xi$	-2.18	$\xi^{-2.18}$
$L_\xi \rightarrow c/H_0$	-7.78	$\xi^{-7.78}$
$\ell_P \rightarrow c/H_0$ (total)	-15.72	$\xi^{-15.72}$

The Geometric Mean

A remarkable finding: L_ξ lies almost exactly at the *geometric mean* of ℓ_P and c/H_0 :

$$\sqrt{\ell_P \cdot \frac{c}{H_0}} = 47.3 \mu\text{m}, \quad L_\xi = 100.2 \mu\text{m}, \quad \frac{L_\xi}{\sqrt{\ell_P \cdot c/H_0}} = 2.12 \quad (9)$$

The Casimir Scale as Cosmic Link

On the ξ -power ladder, the Casimir vacuum scale $L_\xi \approx 100 \mu\text{m}$ lies precisely between the Planck length and the Hubble length. It is neither arbitrary nor accidental, but geometrically determined by the same parameter ξ that generates all the other scales.

.4 Physical Interpretation

Prefactors and Their Meaning

The bridge formula (7) contains two dimensionless factors:

$K_{\text{geo}} = (\pi/2)(15/\pi^2)^{1/4} = 1.744$ A purely geometric factor arising from 3D Stefan–Boltzmann radiation (dimension of phase space) and the torus topology of the T0 time field (factor $\pi/2$, cf. Doc. 159).

$m_e c^2 / (k_B T_{\text{CMB}}) = 2.176 \times 10^9$ The ratio of the electron rest energy to the CMB thermal energy.

On the Decomposition of the Exponent 41/4

The decomposition $41/4 = 10 + 1/4$ has the following meaning:

- The exponent 10 in $\hbar H_0 / (m_e c^2) = (\pi/2) \xi^{10}$ describes how far the cosmological energy scale lies from the particle mass scale – measured in units of ξ .
- The exponent $1/4$ in $L_\xi \propto \xi^{1/4}$ describes how the Casimir scale arises from the thermal length $\hbar c / (k_B T_{\text{CMB}}) = 0.84 \text{ mm}$ – as a fourth root, because the vacuum energy scales as $1/L^4$.
- The sum $41/4$ appears in Doc. 026 as the exponent in $E_H = E_0 \cdot \xi^{41/4}$. The present derivation shows that it can be written as the sum of the exponents of the Casimir relation and the H_0 relation. Since the exponent 10 is calibrated to the measured H_0 , $41/4$ is thereby not derived from ξ either.

Decomposition of the Exponent 41/4

The exponent 41/4 found empirically in Doc. 026 in $E_H = E_0 \cdot \xi^{41/4}$ can be written as the sum:

$$\frac{41}{4} = \underbrace{10}_{\hbar H_0 / (m_e c^2) \sim \xi^{10}} + \underbrace{\frac{1}{4}}_{L_\xi \sim \xi^{1/4} \cdot \hbar c / (k_B T_{\text{CMB}})} \quad (10)$$

The factor $\xi^{1/4}$ attaches to the thermal length $\hbar c / (k_B T_{\text{CMB}}) = 0.84 \text{ mm}$, not to ℓ_P . It encodes the coupling between the vacuum structure (Casimir, 1/4) and the cosmological scale (H_0 , 10). The decomposition is not a derivation from ξ , because the exponent 10 is calibrated.

.5 Experimental Testability

Casimir Precision Measurements

At a plate separation $d = L_\xi \approx 100 \text{ } \mu\text{m}$ the Casimir energy density does not coincide with the CMB energy density but, by Eq. (3), is $\pi^2 / (720\xi) = 102.8$ times larger; equality $\rho_{\text{Casimir}} = \rho_{\text{CMB}}$ occurs at $d = 319 \text{ } \mu\text{m}$. The equation

$$L_\xi^{\text{T0}} = \left(\frac{\xi \hbar c}{\rho_{\text{CMB}}} \right)^{1/4} = 100.24 \text{ } \mu\text{m} \quad (11)$$

is the definition of L_ξ . Because L_ξ is thus determined from the measured CMB energy density and ξ , the ratio of Casimir to CMB energy density at L_ξ is an identity, not measurement evidence. Whether L_ξ has a physical Casimir signature of its own is open.

H_0 Measurement as Indirect Casimir Test

The bridge formula (1) can be solved for H_0 :

$$H_0 = \frac{\pi}{2} \left[\frac{15}{\pi^2} \right]^{1/4} \frac{m_e c^2}{k_B T_{\text{CMB}}} \cdot \frac{c}{L_\xi} \cdot \xi^{41/4} \quad (12)$$

A Casimir experiment could thus determine the Hubble constant indirectly, provided that L_ξ has a physical Casimir signature of its own; this is open. As long as L_ξ is determined from the CMB energy density and the exponent 10 is calibrated, the formula is not an independent prediction of H_0 .

.6 Complete Numerical Chain

Table 3: Numerical verification of the complete chain

Quantity	T0 formula	Numerical value
ξ	$4/30000$	1.333×10^{-4}
L_ξ	$[15\xi/\pi^2]^{1/4} \hbar c/(k_B T)$	$100.24 \mu\text{m}$
ρ_{CMB}	$\xi \hbar c/L_\xi^4$	$4.170 \times 10^{-14} \text{ J/m}^3 \checkmark$
T_{CMB}	Stefan–Boltzmann	$2.725 \text{ K} \checkmark$
$\hbar H_0/(m_e c^2)$	$(\pi/2) \xi^{10}$	2.789×10^{-39}
H_0	from the row above	66.82 km/s/Mpc
$L_\xi \cdot H_0/c$	$K_{\text{geo}} \cdot (m_e c^2/k_B T) \cdot \xi^{41/4}$	$7.241 \times 10^{-31} \checkmark$

Three Independent Inputs – One Consistent System

From the inputs ξ , T_{CMB} and m_e follow the Casimir scale L_ξ and, with the exponent 10 calibrated to H_0 , the Hubble constant H_0 . The bridge formula (1) is satisfied algebraically exactly – no approximation.

.7 Summary

Results of This Document

1. The Casimir scale $L_\xi \approx 100 \mu\text{m}$ and the Hubble constant H_0 are connected by the exact formula $L_\xi \cdot H_0/c = K_{\text{geo}} \cdot (m_e c^2/k_B T_{\text{CMB}}) \cdot \xi^{41/4}$.
2. The exponent $41/4 = 10 + 1/4$ decomposes the value found empirically in Doc. 026: 10 from the H_0 scale (calibrated), $1/4$ from the Casimir scaling.
3. Five scales – sub-Planck, Planck, Compton, Casimir, Hubble – form a ξ -power ladder with exponent steps $\Delta n \approx \{1; 5.77; 2.18; 7.78\}$.
4. The Casimir scale L_ξ lies near the geometric mean of ℓ_P and c/H_0 , with a factor of 2.12.
5. A Casimir precision experiment at $d \approx 100 \mu\text{m}$ could test H_0 indirectly, provided that L_ξ has a Casimir signature of its own; this is open.

Related Documents: Doc. 025/026 (T0 cosmology, $E_H = E_0 \xi^{41/4}$), Doc. 091 (Casimir–CMB connection), Doc. 159 (torus topology), Doc. 165 (H_0 ratio ξ^{10}).